

NATIONAL HACKATHON 2026

RELIEF

Real-time Emergency Logistics & Information Exchange Framework

Coordinating Volunteers & Resources in Disaster Response

TEAM NAME	YuVaRA
MEMBERS	Yugank Rai · Vaibhav Rawat
THEME	Disaster Management
PROBLEM STATEMENT	Omni_DisasterMgmt_2 — Coordinating Volunteers & Resources in Disaster Response

THE DISASTER RESPONSE GAP

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CITIZEN

"Where do I get help?"



VOLUNTEER

"Where am I needed?"



AUTHORITY

"Who is responding?"



RESOURCE MANAGER

"Where are supplies needed?"

THE CORE PROBLEM

- Volunteers do not know where help is needed
- Authorities lack a unified real-time view
- Multiple teams may respond to the same location
- Some zones get excess supplies, others face shortages
- Citizen requests get lost in calls / messages / social media
- Shelters may become overcrowded without visibility

FRAGMENTED INFORMATION

DELAYED COORDINATION

UNEQUAL RESOURCE DISTRIBUTION

ONE PLATFORM. ONE RESPONSE NETWORK.

CITIZEN REQUESTS

RELIEF COORDINATION ENGINE

AUTHORITIES + VOLUNTEERS + RESOURCES + SHELTERS

CORE IDEA



KEY FEATURES

- Unified Incident Dashboard
- Smart Task Matching
- Resource Allocation Tracking
- Shelter Capacity Monitoring
- Verified Emergency Requests
- Real-time Status Tracking

FROM DISTRESS CALL TO RESOLUTION

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1 Citizen reports emergency

2 Request is verified

3 Severity is calculated

4 Nearby suitable volunteer identified

5 Required resource is checked

6 Task is assigned

7 Volunteer responds

8 Status is updated

9 Request is resolved

SYSTEM ARCHITECTURE

USERS	Citizen Volunteer Authority NGO / Resource Manager
FRONTEND	React.js + Tailwind CSS + Leaflet / OpenStreetMap
API LAYER	REST APIs · JWT Authentication · Role-Based Access Control
BACKEND	Node.js + Express.js
CORE SERVICES	Auth · Incident Mgmt · Request Mgmt · Volunteer Matching · Resource Allocation · Shelter Mgmt · Notifications
DATABASE	PostgreSQL(DrizzleORM)
EXTERNAL SERVICES	OpenStreetMap / Leaflet · Email / SMS Notification Service · Cloud Deployment

THE RESPONSE COORDINATION ENGINE

RULE-BASED · NOT AI / ML

1. PRIORITIZE REQUEST

Severity · Waiting Time ·
Vulnerability · Resource Urgency

2. FIND SUITABLE VOLUNTEERS

Required Skill · Distance ·
Availability · Current Workload

3. CHECK RESOURCES

Stock Availability · Nearest Center ·
Demand

4. GENERATE ASSIGNMENT

Highest-Priority Need + Best
Available Match

PRIORITY SCORE

Priority = Severity + Time Waiting + Vulnerability + Resource Urgency

VOLUNTEER MATCHING

Match = Skill + Availability + Distance + Task Priority

EXAMPLE WALKTHROUGH

- Medical emergency reported — high severity
- Nearby verified medical-skill volunteer available
- Medicine stock confirmed at nearest resource point

→ IMMEDIATE ASSIGNMENT

WHAT THE USER ACTUALLY SEES

AUTHORITY DASHBOARD	VOLUNTEER APP	CITIZEN APP	RESOURCE DASHBOARD
Active Incidents: 4	Nearby Tasks (3)	Report Emergency	Inventory Overview
Critical Requests: 12	Priority: High · 1.2 km	Request: Food / Water / Medicine	Demand vs Stock
Volunteers Available: 38	Skill: Medical Assistance	Nearby Shelters	Shortage Alerts
Map + Resource Overview	Accept Task	Track My Request	Allocate to Zone

TECHNOLOGICAL FEASIBILITY

- React + Node.js + PostgreSQL
- REST APIs
- OpenStreetMap for maps
- Cloud deployment (Vercel / Render)
- Role-based JWT authentication

IMPLEMENTATION FEASIBILITY

- MVP buildable with existing web tech
- No hardware / IoT dependency
- Small student team can prototype in hackathon window

KEY CHALLENGES

- Network failures in disaster zones
- False / duplicate reports
- Location accuracy
- Volunteer verification
- Data overload during major events

MITIGATION

- Request verification workflow
- Role-based access control
- Cached essential data (offline)
- Retry / sync mechanism
- Priority queues + admin moderation

IMPACT THAT SCALES



SCALE PROGRESSION



FUTURE INTEGRATIONS (PROPOSED / FUTURE SCOPE)



RESEARCH AREAS

- Disaster response coordination models
- Volunteer management frameworks
- Humanitarian logistics & supply chains
- Emergency resource allocation strategies
- Crisis mapping methodologies
- Shelter management best practices

REFERENCE SOURCES

- NDMA — National Disaster Management Authority
- UNDRR — UN Office for Disaster Risk Reduction
- WHO — Emergency & Health Guidelines
- IFRC — International Federation of Red Cross
- OpenStreetMap official documentation

RELIEF turns fragmented disaster response into coordinated action.

NEED

PEOPLE

RESOURCES

RESPONSE